

Within-family consistency and the descriptive fit-matrix

Of 13 evaluated family × task cells, 12 agree; the lone within-family split is the T2 donor OT pair.

DESCRIPTIVE / EXPLORATORY

PREREG rule 5, not a hypothesis test

a Within-family verdict agreement

cell value: ρ = within-family per-unit Spearman concordance
CONTEXT-TRANSFER

	CONTEXT-TRANSFER			UNSEEN-PERTURBATION		
	T1 cell-context	T2 donor	T5 cell-context	T4 modality	T3 unseen-pert	T5 unseen-cpd
Latent	none beat $\rho=+0.48$	none beat $\rho=+0.44$	none beat $\rho=-0.80$	none beat PENDING $n<3$ units	none beat $\rho=+0.70$	none beat ρ n.a.
Graph				none beat PENDING $n<3$ units	none beat $\rho=+0.90$	
Foundation					none beat $\rho=+1.00$	
Hybrid					none beat $\rho=+0.70$	
Chemistry						none beat ρ n.a.
OT		SPLIT CellOT beats scPRAM fails		none beat PENDING single-seed		

OT-family donor exception

the lone donor win is
CellOT-specific, not family-level

CellOT 0.369

scPRAM 0.159

paired $n = 106$ shared donors
scPRAM wins 1/106 gap -0.210
Wilcoxon $p = 4.3 \times 10^{-19}$

b Descriptive fit-matrix: does the family beat the universal floor?

cell value: best-member gap vs universal floor (CI_lo = cluster-bootstrap lower bound)
CONTEXT-TRANSFER

	CONTEXT-TRANSFER			UNSEEN-PERTURBATION		
	T1 cell-context	T2 donor	T5 cell-context	T4 modality	T3 unseen-pert	T5 unseen-cpd
Latent	-0.014 $CI_lo -0.048$	-0.067 $CI_lo -0.077$	-0.092 $CI_lo -0.153$	-0.112 PENDING $n<3$ units	-0.379 $CI_lo -0.544$	
Graph				-0.150 PENDING $n<3$ units	-0.266 $CI_lo -0.316$	
Foundation					-0.436 $CI_lo -0.501$	
Hybrid		-0.089 $CI_lo -0.101$	-0.214 $CI_lo -0.261$	-0.636 PENDING $n<3$ units	-0.479 $CI_lo -0.542$	
Chemistry			+0.119 $CI_lo +0.101$			-0.008 PENDING $n<3$ units
Shift				-0.044 PENDING $n<3$ units		

beats floor ($CI_low > 0$) consistent: none beat within-family split pending ($n<3$ / single-seed) not evaluated